

Q1. Write a member function for Simple Binary Tree to return sum of all nodes at level K?

```
int sumNodesAtK(BTNode* t, int level){ //initially root is passed as first parameter of the function
    if (!t) return 0;
    if (!k) return t -> val;
    int s = sum(t -> left, k - 1);
    s += sum(t -> right, k - 1);
    return s;
}
```

Q2. Write a member function for binary search tree, to count and print number of elements larger than the target value.

```
int printLargerThanK(BTNode* t, const int K){
    //initially root is passed as first parameter of the function
    if (!t) return 0;
    int count = 0;
    if (t -> val > K){
        count++;
        count += printLargerThanK(t -> left, K);
    }
    count += printLargerThanK(t -> right, K);
    return count;
}
```

Q3. Create an AVL tree with height = 6, having minimum number of nodes. The definition of height is "Number of nodes on the longest path".

The tree is an AVL and has six nodes in the right most branch.

To the best of my observation, if I remove any node the tree will rotate and height will reduce from 6 to 5.

Therefore, this is the AVL of height = 6, with minimum number of nodes.

